

What is a Transistor?

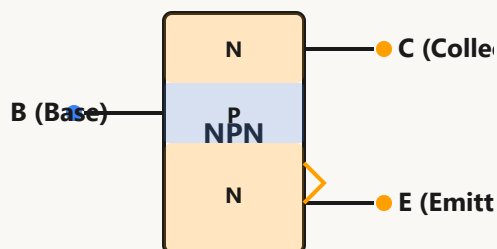
1 What is a Transistor?

A **transistor** is a tiny electronic component that acts like a **switch** or an **amplifier**. It's the fundamental building block of all modern electronics — computers, phones, radios, industrial controllers, everything.

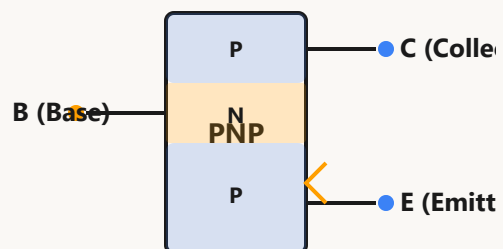
Think of it as a **tap**: a small turn of the handle (base current) controls a huge flow of water (collector-emitter current).

Invented in 1947 at Bell Labs, the transistor replaced bulky vacuum tubes and made electronics small, efficient, and reliable. A modern CPU contains **billions** of transistors on a chip smaller than your thumbnail.

NPN TRANSISTOR



PNP TRANSISTOR



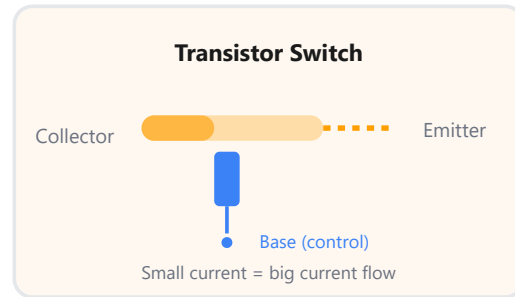
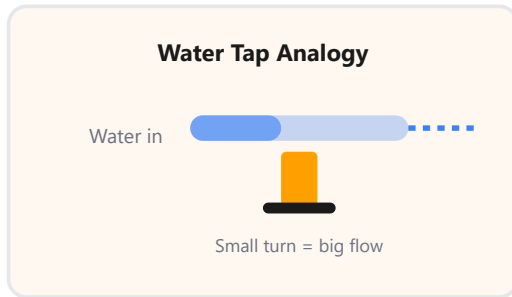
2 How Does It Work?

A transistor has **3 terminals**: **Base (B)**, **Collector (C)**, and **Emitter (E)**.

In an **NPN** transistor (most common):

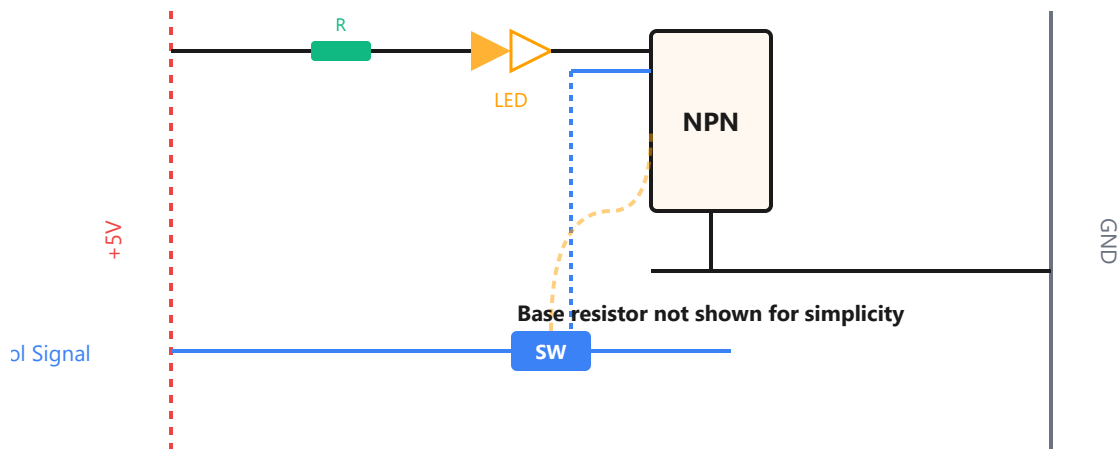
- A small current flowing from **Base** → **Emitter** turns the transistor ON
- This allows a much larger current to flow from **Collector** → **Emitter**

- No base current = no collector current = transistor is OFF
- A small input controls a big output — that's **amplification**!



3 Transistor as a Switch (Most Common Use)

This is how transistors are used in industrial electronics — to turn things ON and OFF based on a control signal.



4 How a Transistor Thinks



Base (B) — The Gatekeeper

A tiny signal here decides if the big current flows or not. No signal = OFF.



Collector (C) — The Power In

This is where the main current enters. Think of it as the water pipe inlet.



Emitter (E) — The Power Out

The current exits here. When Base is ON, Collector→Emitter flows freely.



Gain (hFE) — Amplification Factor

A typical transistor has gain of 100-800. 1mA at Base can control 100mA at Collector!

5 Real-World PCB Examples

Here's where you'll find transistors in industrial boards:

- **Power supply boards** — Switching transistors in SMPS circuits (MOSFETs)
- **Relay drivers** — A transistor amplifies a PLC signal to energize a relay coil
- **Sensor inputs** — Transistor buffers protect the controller from voltage spikes
- **Motor speed control** — Power transistors (IGBTs) in VFD drives
- **Signal amplifiers** — Amplify weak thermocouple or RTD signals

When repairing PCBs, the most common transistor faults are: shorted C-E (stuck ON), open B-E junction (stuck OFF), or leaky C-E (partial ON). All easily checked with a multimeter.

Quick Quiz

In an NPN transistor, a small current at which terminal controls the large current flow?

A) Collector

B) Base

C) Emitter

D) Substrate